

Homework 9

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Example 9.2: Bayesian Probit Model

1. Background

In a medical study on infections resulting from birth by Caesarean section *three* influence factors have been studied:

- z_{i1} : an indicator of whether the Caesarian was planned or not
- z_{i2} : an indicator of whether additional risk factors were present at the time of birth
- z_{i3} : an indicator of whether antibiotics were given as a prophylaxis.

The response Y_i is the number of **infections** that were observed amongst n_i patients having the same influence factors (covariates).

```
y <- c(11, 1, 0, 23, 28, 0, 8)      # Number of births with infection
n <- c(98, 18, 2, 26, 58, 9, 40)    # total
z1 <- c(1, 0, 0, 1, 0, 1, 0)        # planned
z2 <- c(1, 1, 0, 1, 1, 0, 0)        # risk factors
z3 <- c(1, 1, 1, 0, 0, 0, 0)        # antibiotics
Z <- cbind(1, z1, z2, z3)
```

2. Model

The data can be modeled by assuming that:

$$Y_i \sim \text{Binomial}(n_i, \Phi(z_i' \beta))$$

Where:

- $z_i = (1, z_{i1}, z_{i2}, z_{i3})$
- $\Phi(\cdot)$ is the CDF of the standard normal distribution $N(0, 1)$.

A suitable normal prior distribution is chosen for the parameter of interest β :

$$\beta \sim N(0, \mathbb{I}/\lambda)$$

According to Bayes' theorem, the posterior density $f(\beta|y_1, \dots, y_n)$ is:

$$f(\beta|y_1, \dots, y_n) \propto \left(\prod_{i=1}^N \Phi(z'_i \beta)^{y_i} \cdot (1 - \Phi(z'_i \beta))^{n_i - y_i} \right) \cdot \exp \left(-\frac{\lambda}{2} \sum_{j=0}^3 \beta_j^2 \right)$$

```
# posterior density
log_posterior <- function(beta, y, n, Z, lambda = 0) {
  eta <- as.vector(Z %*% beta) # z_i' * beta
  p <- pnorm(eta) # 计算标准正态分布的累积分布函数 Phi(z_i' * beta)
  p <- pmax(pmin(p, 1 - 1e-10), 1e-10) # 避免 p=0 或 p=1
  log_lik <- sum(y * log(p) + (n - y) * log(1 - p))
  log_prior <- - (lambda / 2) * sum(beta^2)
  return(log_lik + log_prior)
}
```

3. The random walk Metropolis algorithm

Starting from an initial value $\beta^{(0)}$, generate posterior samples by the following random walk Metropolis steps for $t = 1, 2, \dots$:

1. Draw $\epsilon \sim N(0, \Sigma)$ and propose

$$\beta = \beta^{(t-1)} + \epsilon.$$

2. Compute

$$\alpha(\beta \mid \beta^{(t-1)}) = \min \left\{ 1, \frac{f(\beta \mid Y_1, \dots, Y_n)}{f(\beta^{(t-1)} \mid Y_1, \dots, Y_n)} \right\}.$$

3. With probability $\alpha(\beta \mid \beta^{(t-1)})$, accept the proposal and set $\beta^{(t)} = \beta$; otherwise keep the previous value $\beta^{(t)} = \beta^{(t-1)}$.

Here the proposal covariance is chosen as $\Sigma = 0.08I$. We run 50,000 iterations and discard the first 10,000 samples as burn-in. The results are similar to those obtained in the lecture notes.

```
n_iter <- 50000      # 总次数
burn_in <- 10000     # 丢弃前 10000 次
```

```

run_rwm <- function(Sigma, n_iter, burn_in, y, n, Z, beta_init = rep(0, 4), seed = 42) {
  set.seed(seed)
  beta_chain <- matrix(0, nrow = n_iter, ncol = length(beta_init))
  beta_current <- beta_init
  accept_count <- 0

  for (t in 1:n_iter) {
    beta_prop <- mvrnorm(1, mu = beta_current, Sigma = Sigma) # 生成候选值
    log_p_prop <- log_posterior(beta_prop, y, n, Z)
    log_p_curr <- log_posterior(beta_current, y, n, Z)
    log_alpha <- log_p_prop - log_p_curr # 接受概率

    if (log(runif(1)) < log_alpha) {
      beta_current <- beta_prop
      accept_count <- accept_count + 1
    } # 接受的情况

    beta_chain[t, ] <- beta_current
  }

  beta_samples <- beta_chain[(burn_in + 1):n_iter, ] # 丢弃 burn in

  list(
    beta_chain = beta_chain,
    beta_samples = beta_samples,
    post_mean = colMeans(beta_samples),
    ci_lower = apply(beta_samples, 2, quantile, probs = 0.025),
    ci_upper = apply(beta_samples, 2, quantile, probs = 0.975),
    accept_count = accept_count,
    acceptance_rate = accept_count / n_iter
  )
}

Sigma <- 0.08 * diag(4) #
rwm_result <- run_rwm(Sigma, n_iter, burn_in, y, n, Z)
beta_chain <- rwm_result$beta_chain
beta_samples <- rwm_result$beta_samples
post_mean <- rwm_result$post_mean
ci_lower <- rwm_result$ci_lower

```

```
ci_upper <- rwm_result$ci_upper
accept_count <- rwm_result$accept_count
```

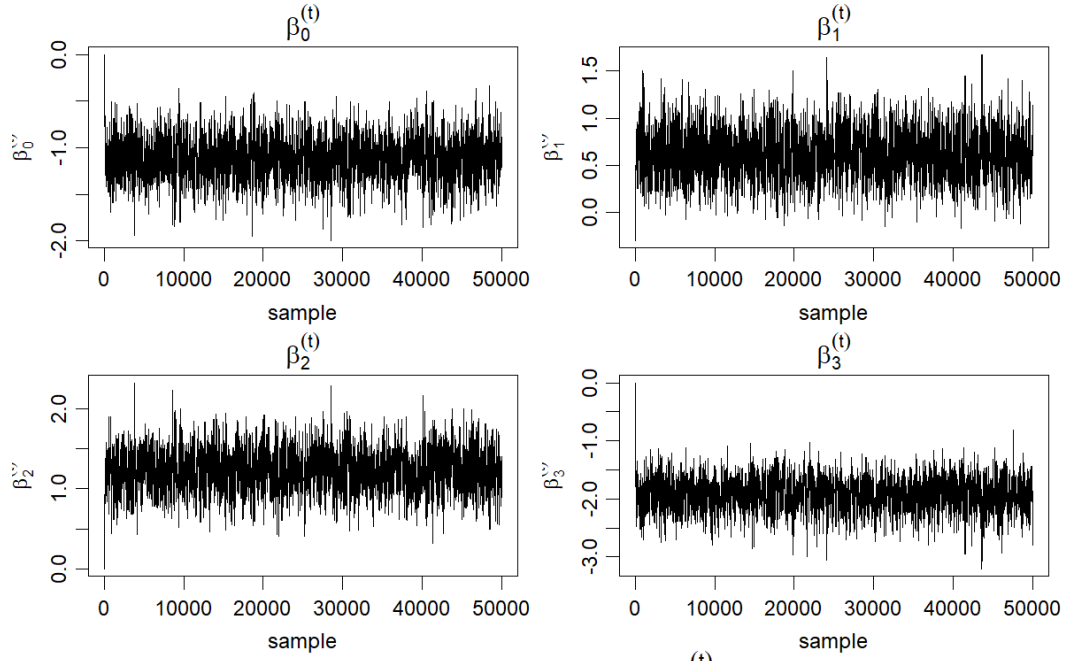
```
# 接受率
cat("Acceptance rate:", rwm_result$acceptance_rate * 100, "%\n\n")
Acceptance rate: 13.916 %
```

```
correlation1 <- numeric(4)
names(correlation1) <- paste0("beta", 0:3)

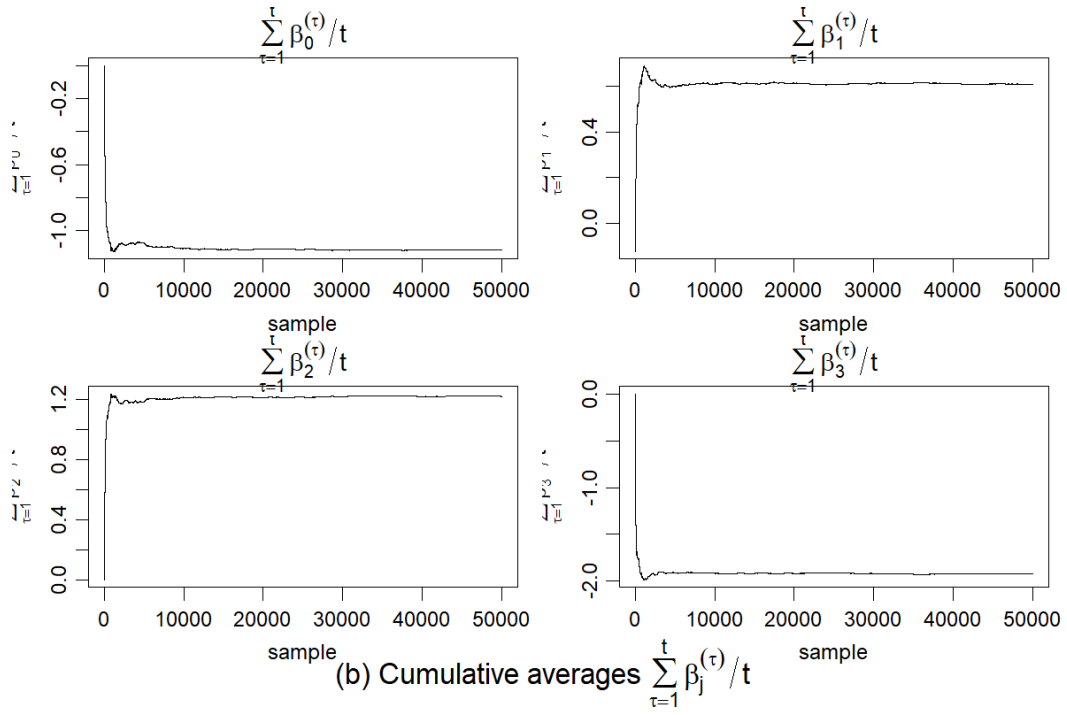
for (j in 1:4) {
  correlation1[j] <- cor(
    beta_samples[-nrow(beta_samples), j],
    beta_samples[-1, j]
  )
}
correlation1 # autocorrelation
      beta0      beta1      beta2      beta3
0.9455465 0.9477525 0.9520974 0.9516373
```

```
results <- data.frame(
  Parameter = c("intercept (beta0)", "planned (beta1)", "risk factors (beta2)", "antibiotics (beta3)"),
  Posterior_Mean = round(post_mean, 4),
  CI_2.5 = round(ci_lower, 4),
  CI_97.5 = round(ci_upper, 4)
)
print(results) # 后验均值和 95% 置信区间
```

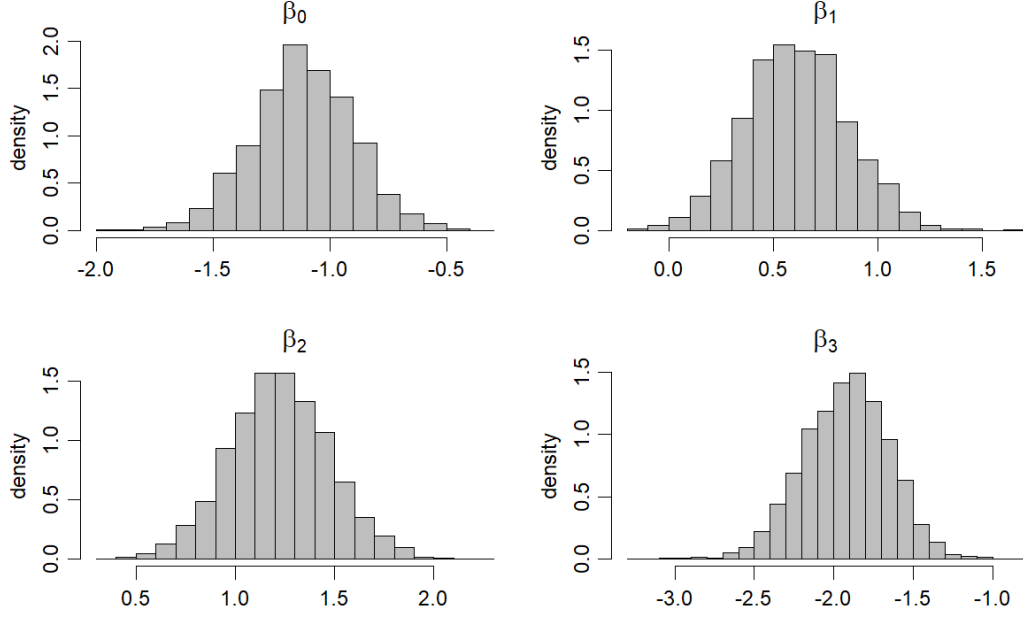
	Parameter	Posterior_Mean	CI_2.5	CI_97.5
1	intercept (beta0)	-1.1155	-1.5510	-0.6955
2	planned (beta1)	0.6094	0.1348	1.0900
3	risk factors (beta2)	1.2218	0.7338	1.7291
4	antibiotics (beta3)	-1.9220	-2.4575	-1.4207



(a) Sample paths of the $\beta_j^{(t)}$



(b) Cumulative averages $\sum_{\tau=1}^t \beta_j^{(\tau)} / t$



(c) Posterior distributions of the β_j

4. A New Proposal Distribution

We can use the idea of GLM to approximate the posterior covariance.

Starting with the log-likelihood for the binomial response Y_i :

$$\log L_i = Y_i \log(p_i) + (n_i - Y_i) \log(1 - p_i)$$

Taking the derivative with respect to β using the chain rule:

$$U_i(\beta) = \frac{\partial \log L_i}{\partial \beta} = \frac{\partial \log L_i}{\partial p_i} \cdot \frac{\partial p_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta}$$

Evaluating these derivatives, we obtain:

$$U_i(\beta) = \frac{Y_i - n_i p_i}{p_i(1 - p_i)} \cdot \phi(\eta_i) \cdot z_i$$

The Fisher information matrix for the i -th observation is:

$$\mathcal{J}_i(\beta) = \text{Var}(U_i(\beta)) = \left(n_i \frac{\phi(\eta_i)^2}{p_i(1 - p_i)} \right) z_i z_i'$$

Summing over all N observations gives the total Fisher information matrix:

$$\mathcal{J}(\beta) = \sum_{i=1}^N \mathcal{J}_i(\beta) = \sum_{i=1}^N \underbrace{\left(n_i \frac{\phi(\eta_i)^2}{p_i(1 - p_i)} \right)}_{d_i} z_i z_i'$$

where d_i defines the elements of a diagonal weight matrix D :

$$d_i = n_i \frac{\phi(\eta_i)^2}{p_i(1-p_i)}$$

Therefore, we choose the new proposal covariance as

$$\Sigma_{\text{new}} = (Z'DZ)^{-1}$$

where Z is the design matrix and D is the diagonal weight matrix from the GLM approximation.

Note: To ensure a fair comparison, we scale Σ_{new} so that its determinant matches that of the original Σ .

But we didn't use $\Sigma_{\text{new}} = 2(Z'DZ)^{-1}$ as in the lecture notes

```
probit_glm <- glm(cbind(y, n - y) ~ z1 + z2 + z3, family = binomial(link = "probit"))
eta_hat <- as.vector(Z %*% coef(probit_glm))
p_hat <- pnorm(eta_hat)
phi_hat <- dnorm(eta_hat)
D <- diag(as.vector(n * phi_hat^2 / (p_hat * (1 - p_hat))))
Sigma_new <- solve(t(Z) %*% D %*% Z)
scale_factor <- (det(Sigma) / det(Sigma_new))^(1 / 4) # Scale to match determinants
Sigma_new <- scale_factor * Sigma_new
round(Sigma_new, 4) # 新的 Sigma
      z1      z2      z3
0.1168 -0.0335 -0.1076  0.0215
z1 -0.0335  0.1381 -0.0041 -0.0891
z2 -0.1076 -0.0041  0.1550 -0.0436
z3  0.0215 -0.0891 -0.0436  0.1627
```

```
rwm_result_new <- run_rwm(Sigma_new, n_iter, burn_in, y, n, Z)
beta_chain_new <- rwm_result_new$beta_chain
beta_samples_new <- rwm_result_new$beta_samples
post_mean_new <- rwm_result_new$post_mean
ci_lower_new <- rwm_result_new$ci_lower
ci_upper_new <- rwm_result_new$ci_upper
accept_count_new <- rwm_result_new$accept_count
# 新的接受率
cat("Acceptance rate with new Sigma:", rwm_result_new$acceptance_rate * 100, "%\n\n")
Acceptance rate with new Sigma: 20.23 %
```

```

correlation_new <- numeric(4)
names(correlation_new) <- paste0("beta", 0:3)

for (j in 1:4) {
  correlation_new[j] <- cor(
    beta_samples_new[-nrow(beta_samples_new), j],
    beta_samples_new[-1, j]
  )
}

correlation_new # autocorrelation with new Sigma
  beta0      beta1      beta2      beta3
0.8653673 0.8708656 0.8698870 0.8700226

```

表 1: Comparison for two proposal distributions

Proposal	Acceptance_Rate	beta0	beta1	beta2	beta3
$\Sigma = 0.08 \times I$	0.1392	0.9455	0.9478	0.9521	0.9516
$\Sigma = 2(Z'DZ)^{-1}$	0.2023	0.8654	0.8709	0.8699	0.8700

Conclusion

As the lecture notes said,

The autocorrelation is reduced, and the proportion of accepted values obtained increases to 20.23 %. Note that the determinant of both choices of Σ was chosen to be the same, so the improvement of the mixing behaviour is due to a difference in the structure of the covariance.